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Lunar soil characterization consortium analyses: Pyroxene and maturity estimates derived from Clementine image data

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Abstract

The mineralogy of a planetary surface is a diagnostic product of its formation and geologic evolution. Global assessment of lunar mineralogy at high spatial resolution has been a long standing goal of lunar exploration. Currently, the only global data available for such study is multispectral imagery from the Clementine mission. We use the detailed compositional, petrographic, and spectroscopic data of lunar soils produced by the Lunar Soil Characterization Consortium to explore the use of multispectral imaging as a diagnostic tool. We compare several statistically optimized formulations of links between spectral and mineral parameters and apply them to Clementine UV–VIS data. The most reliable results are for estimations of pyroxene abundance and maturity parameters (agglutinate abundance, I_s/FeO). Estimations of different pyroxene composition (low-Ca versus high-Ca) appear good in a relative sense, but absolute values are limited by residual wavelength dependent Clementine photometric calibrations. Since the signal-to-noise of Clementine multispectral data is good at the 1-km scale, almost any combination of parameters that capture inherent spectral variance can provide spatially coherent maps, although the parameters may not actually be directly related to composition. Clementine estimates are useful for identifying scientific or exploration targets for imaging spectrometer sensors of the next generation that are specifically designed to characterize mineralogy.
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1. Introduction

Understanding the Moon requires an understanding of its mineralogy, the aggregate of which makes up its rocks. For example, the magma ocean hypothesis of the origin and evolution of the crust/mantle relies on the distribution and relative abundance of mafic minerals and plagioclase. The basaltic nature of the lunar maria, and specifically the abundance of high-Ca pyroxene and TiO_2 content, is a constraint on the character of the lunar mantle from which they were derived. It is well known that the spectral properties of lunar materials are directly linked to their mineralogy (Adams and McCord, 1971a, 1971b, 1972, 1973), and it has been a long-term goal to use this association to assess the mineralogy of the Moon with remote sensors (McCord and Adams, 1973; McCord et al., 1981).

Interpretation of spectral properties of lunar materials relies on fundamental characteristics of diagnostic absorptions that are based on principals derived from mineral physics (e.g., Burns, 1993). Such applications, however, require high precision visible to near-infrared spectra (0.4–2.6 μm) of high spectral resolution, often also acquired at high spatial resolution. Spectroscopic analysis has been highly successful for single targets (3–20 km in diameter) on the lunar nearside using instruments developed in the late 1970's for use on Earth-based telescopes (McCord et al., 1981; Pieters, 1986, 1993; Hawke et al., 2003). In contrast, the majority of optical data currently available for the Moon comes from much simpler sensors, namely multispectral imagers, which typically consist of a digital framing camera equipped with several filters. Multispectral imaging has the advantage of extensive two-dimensional spatial cover-

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age, but with the loss of spectral resolution and range. The state of CCD silicon detector technology limited the spectral range of such systems to be between ~350 to 1000 nm, with most orbital systems containing 3–6 filters (bandpasses) within that range. Examples for lunar applications are the Galileo SSI camera (6 broadband filters), the UV–VIS camera of Clementine (5 filters), and the AMIE camera of ESA Smart-1 (3 filters).

We previously investigated relationships between the general spectral properties of lunar samples for mare soils and their measured compositional characteristics (Pieters et al., 2002). We initially used a statistical analysis of nine mare soil samples of different composition and maturity. To probe information in high spectral resolution spectra of the lunar samples obtained in the laboratory over the range 0.35–2.50 μm, a principal-component analysis was used to develop a statistical link between spectral properties and compositional parameters. Unfortunately, such an approach cannot be tested and applied directly to map chemical and mineral composition of the lunar surface since only a limited number of individual high spectral resolution spectra exist for Earth-based measurements (e.g., McCord et al., 1981; Pieters and Pratt, 2000). We later explored use of the spectral bands of Clementine UV–VIS and AMIE/Smart-1 cameras (Shkuratov et al., 2003a, 2003b) to investigate relations between more limited spectral and compositional parameters derived from the same set of mare soil samples (Pieters et al., 2002). A key aspect of these data is that Clementine data and laboratory spectral measurements of lunar soils are tied to the same photometric geometry and calibration (Pieters, 1999).

2. Objectives and data

The analysis presented here further explores the strengths and limits of using multispectral imaging to investigate compositional parameters across the Moon. The data are summarized in Table 1 and discussed below. For these new analyses, we use laboratory data that span a much wider suite of lunar samples, including the original Lunar Soil Characterization Consortium (LSCC) mare soils as well as additional highland soils from the Apollo 14 and 16 landing sites (Taylor et al., 2001, 2003, in preparation). The LSCC selected this broad suite of soils to be representative of both the compositional diversity found in lunar soils as well as the degree of "maturity" or exposure to the space environment. Both high- and low-Ti mare soils were included and very "immature" as well as "mature" soils from both mare and highland sites. The soils are representative of Apollo landing sites, but, of course, do not include examples of unsampled soils known to cover large areas of the Moon such as the young high-Ti basalts of the western nearside (e.g., Pieters, 1978) that are believed to be olivine-rich (Staid and Pieters, 2001).

The LSCC carried out detailed measurements of elemental and mineral composition for three size fractions of each soil along with measurements of I_s/FeO, a measure of maturity (Morris, 1976, 1978). Since the optical properties of lunar soils are controlled by the smaller size fractions (Pieters et al., 1993) and space weathering processes (Pieters et al., 2000), the LSCC concentrated on detailed analysis of the <45 μm components of lunar soils. In the detailed compositional analysis of each subsample, pyroxenes were assessed in four compositional classes and the proportion of each was determined (e.g., see Taylor et al., 2001, 2003). High spectral resolution bi-directional spectra were acquired for each subsample in the RELAB at Brown University. All spectra were measured over the spectral range 0.3–2.6 μm with 5 nm sampling resolution at a phase angle of 30° ($i = 30°$; $e = 0°$, the angles of incidence and emergency, respectively) (Pieters and Hiroi, 2004). Examples of LSCC spectra are shown in Fig. 1. All data (spectra and composition) are available on the web at http://www.planetary.brown.edu/relabdocs/LSCCsoil.html.

The carefully coordinated mineral and spectral investigations produced by the LSCC allow detailed analyses of the correlation between important parameters of lunar soils (e.g., mineral content) and their spectral properties. Lunar soil samples returned from all of the Apollo missions are included in the LSCC suite. Each soil was subdivided into three size fractions (<10, 10–20, and 20–45 μm) in addition to a "bulk" <45 μm sample (see Taylor et al., 2001, for description). Altogether we analyzed 52 samples of the different size fractions and these are listed in Table 1. This set includes more than twice as many samples than our previous studies that were limited to basalt soils (Pieters et al., 2002; Shkuratov et al., 2003a). Since the Clementine data are the only global spectral data for the Moon, we re-sampled our high spectral resolution spectra of soils from RELAB to the five bands of the UV–VIS camera.

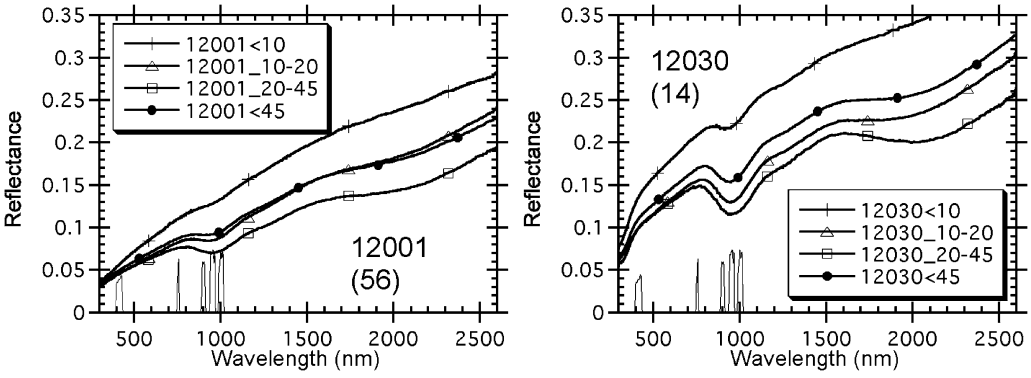


Fig. 1. Example of bidirectional reflectance spectra of LSCC soils and their size separates. Both are from the basaltic Apollo 12 site. Scaled Clementine UV–VIS bandpasses are shown at the bottom. The number in parentheses is a measure of the maturity parameter I_s/FeO for the <250 μm bulk soil.

Table 1
Spectral and compositional data for lunar soils used in this analysis

Soil	Size (μm)	*A* (415) (%)	*A* (753) (%)	*A* (899) (%)	*A* (952) (%)	*A* (1000) (%)	I_s	I_s/FeO	Agg. (%)	Augite (%)	Fe-Px (%)	Pigeo (%)	OPx (%)	Total Px (%)
10084	<10	4.61	8.00	9.04	9.31	9.62	1772	145	62.57	5.96	0.25	1.81	0.42	8.44
10084	10–20	4.32	6.91	7.46	7.51	7.68	1318	88	57.02	7.81	0.57	3.23	0.61	12.22
10084	20–45	4.37	6.45	6.40	6.26	6.29	1055	67	53.87	10.35	1.38	3.94	0.34	16.01
10084	Bulk	4.49	7.28	7.88	7.95	8.14	1302	88						
12001	<10	5.63	10.98	12.41	12.80	13.32	1465	115	61.86	4.79	0.58	6.06	2.07	13.5
12001	20–45	4.64	7.59	7.20	7.04	7.26	876	51	56.15	7.49	1.76	8.74	1.82	19.81
12001	Bulk	4.95	8.71	9.11	9.19	9.54	992	62						
12030	<10	13.53	20.85	21.69	21.82	22.66	466	32	54.95	5.18	0.89	7.07	2.2	15.34
12030	10–20	9.91	15.44	13.51	13.00	13.64	296	17	49.77	6.81	1.79	9.93	2.88	21.42
12030	20–45	9.89	14.76	12.15	11.57	11.98	214	12	39.39	12.14	2.59	15.2	3.85	33.78
12030	Bulk	10.98	16.88	15.73	15.42	16.09	320	20						
14141	<10	21.68	32.96	33.58	33.77	34.95	111	14.5	45.86	2.19	0.44	4.29	3.37	10.29
14141	10–20	16.89	25.90	24.00	23.90	25.72	110	11.6	48.63	1.85	0.38	4.58	4.07	10.88
14141	20–45	13.58	20.15	16.03	15.73	17.35	67	5.8	40.96	3.08	1.08	8.08	7.57	19.81
14141	Bulk	17.67	26.27	24.52	24.47	26.05	95	9.7						
14163	<10	11.27	19.80	21.79	22.43	23.34	768	87	66.33	0.91	0.14	1.38	1.51	3.94
14163	10–20	7.99	13.93	14.38	14.64	15.47	654	64.8	58.46	2.41	0.78	4.94	5.68	13.81
14163	20–45	6.09	10.22	9.42	9.44	10.07	497	43.2	56.44	3.1	0.92	5.66	6.5	16.18
14163	Bulk	9.01	15.42	16.30	16.72	17.50	661	66.5						
14259	<10	7.62	15.75	17.94	18.70	19.55	1367	174.8	71.58	1.77	0.29	1.96	1.92	5.94
14259	10–20	5.92	11.85	13.01	13.47	14.26	988	101.8	68.73	1.66	0.5	3.18	3.72	9.06
14259	20–45	4.87	8.09	8.21	8.27	8.77	849	77.2	60.53	3.04	1.57	6.14	7.4	18.15
14259	Bulk	6.04	11.47	12.56	12.95	13.63	1036	108.6						
14260	<10	6.55	13.90	15.91	16.54	17.29	1174	144.9	66.47	1.51	0.48	3.15	2.58	7.72
14260	10–20	5.70	11.86	12.92	13.33	14.08	973	98.9	65.19	2	0.64	4.33	5.14	12.11
14260	20–45	4.63	7.94	7.90	8.03	8.53	858	80.2	64.04	3.07	0.94	4.99	4.68	13.68
14260	Bulk	5.71	11.12	12.11	12.47	13.15	900	93.3						
15041	<10	5.33	11.30	12.76	13.25	13.85	1802	161	70.27	1.62	0.42	2.49	0.79	5.32
15041	10–20	4.18	7.63	7.91	7.95	8.33	1344	92	56.68	5.12	1.37	8.14	2.35	16.98
15041	20–45	4.58	7.55	7.28	7.33	7.64	1020	66	51.28	6.75	1.49	10.48	3.77	22.48
15041	Bulk	4.61	8.67	9.16	9.35	9.74	1321	93						
15071	10–20	4.73	8.78	8.80	8.80	9.24	1248	80	49.17	5.56	1.38	7.64	2.13	16.71
15071	20–45	4.95	8.51	8.00	7.99	8.37	774	49	47.63	6.98	1.6	10.27	3.22	22.07
15071	Bulk	4.99	9.60	10.06	10.24	10.70	1058	71						
61141	<10	13.52	24.84	27.51	28.39	29.32	437	119.3	62.63	0.06	0.19	0.23	0.22	0.7
61141	10–20	10.28	18.21	19.60	20.11	20.97	419	81.6	49.83	0.04	1.45	2.15	1.69	5.33
61141	20–45	7.59	12.70	13.51	13.87	14.51	389	75.5	50.08	0.18	1.11	1.38	1.68	4.35
61141	Bulk	11.34	19.66	21.33	21.96	22.79	454	94.5						
61221	<10	26.95	35.21	36.11	36.36	36.86	123	27	41.59	0.37	0.02	0.55	0.56	1.5
61221	10–20	24.13	32.61	31.37	31.23	32.05	55	12	32.56	0.14	1.95	1.43	1.82	5.34
61221	20–45	23.13	30.03	26.75	26.26	27.02	46	10	28.92	0.19	1.98	2.24	2.96	7.37
62231	<10	13.25	23.69	26.49	27.31	28.22	613	169	69.48	0.3	0.03	0.27	0.28	0.88
62231	10–20	11.48	19.55	21.15	21.67	22.44	534	109.9	51.02	1.74	0.12	1.55	1.99	5.4
62231	20–45	10.39	14.95	15.93	16.32	16.86	429	80.7	50.57	1.52	0.19	1.33	2.08	5.12
62231	Bulk	11.06	18.65	20.35	20.95	21.75	568	116.7						
64801	<10	13.93	25.67	28.61	29.54	30.53	442	115.2	62.57	0.64	0.003	0.84	1.18	2.663
64801	10–20	11.17	20.20	22.00	22.66	23.55	406	84.9	61	0.6	0.005	0.96	1.24	2.805
64801	20–45	10.19	17.53	18.65	19.14	19.87	402	83.4	53.59	1.33	0.01	1.15	2.03	4.52
64801	Bulk	11.35	19.68	21.45	22.07	22.92	431	92.2						
67461	<10	27.71	41.59	43.06	43.64	44.95	118	35.2	31.88	0.04	1.06	0.64	1.09	2.83
67461	10–20	22.91	33.95	33.45	33.90	35.48	111	23.9	32.42	0.05	1.52	1.07	1.47	4.11
67461	20–45	19.43	28.13	27.76	28.14	29.42	110	22.3	25.36	0.18	2.53	1.61	2.96	7.28
67461	Bulk	25.04	36.15	36.08	36.69	38.29	126	29.8						
67481	<10	25.68	39.33	41.46	42.31	43.65	139	38.5	35.09	0.05	1.41	1.05	1.38	3.89
67481	10–20	21.30	32.12	32.65	33.21	34.66	133	33	28.54	0.13	1.73	1.27	2.55	5.68
67481	20–45	16.41	23.55	23.68	24.02	25.05	107	20.7	27.57	0.17	1.94	1.54	2.95	6.6
67481	Bulk	22.80	32.61	33.47	34.18	35.47	147	33.5						
70181	<10	5.57	10.20	11.83	12.32	12.79	1345	104	58.3	1.98	0.31	1.76	0.59	4.6
70181	10–20	5.14	8.04	8.76	8.94	9.24	995	63	51.7	3.74	0.97	2.57	1.2	8.5
70181	20–45	4.73	7.06	7.23	7.24	7.44	865	53	43.4	8.15	1.37	4.7	1.51	15.7
70181	Bulk	4.88	7.84	8.59	8.79	9.09	933	61						
71501	<10	4.62	8.51	9.91	10.22	10.55	1212	88	53.1	4.1	0.76	2.85	1	8.8

(*continued on next page*)

Table 1 (*continued*)

Soil	Size (μm)	*A* (415) (%)	*A* (753) (%)	*A* (899) (%)	*A* (952) (%)	*A* (1000) (%)	I_s	I_s/FeO	Agg. (%)	Augite (%)	Fe-Px (%)	Pigeo (%)	OPx (%)	Total Px (%)
71501	10–20	4.60	7.55	8.11	8.11	8.33	838	50	44.8	6.34	1.25	4.61	1.47	13.7
71501	20–45	4.66	7.51	7.37	7.16	7.29	508	28	38.3	11.1	2.35	6.31	1.44	21.3
71501	Bulk	4.63	7.70	8.37	8.46	8.69	726	44						
79221	<10	5.19	9.06	10.45	10.89	11.36	1949	169	61.5	1	0.6	1.42	0.6	3.6
79221	10–20	4.28	6.76	7.06	7.10	7.43	1192	78	54.3	3.24	1.82	2.86	1.64	9.7
79221	20–45	4.68	6.89	6.94	6.98	7.15	921	57	46.5	4.85	3.14	3.72	1.47	13.5
79221	Bulk	4.74	7.63	8.51	8.75	9.06	1274	91						

Mare soils are 10*xxx*, 12*xxx*, 15*xxx*, and 7*xxxx* for Apollo 11, 12, 15, and 17, respectively. Apollo 14 soils are 14*xxx*; Apollo 16 highland soils are 6*xxxx*. *A* (*nnn*) refers to Clementine albedo of the samples at band *nnn*. I_s and I_s/FeO are parameters linked to maturity (see text); agg. = agglutinate; Fe-Px = iron-rich pyroxene; pigeo = pigeonite; OPx = orthopyroxene; total Px = total pyroxene.

The re-sampled soil data are listed in Table 1 and are used in this statistical analysis linking spectral properties to compositional and maturity parameters. To evaluate the lunar soil characteristics on a global scale, we used the 1-km UV–VIS Clementine mosaics (Eliason et al., 1999). These mosaics correspond to five spectral bands centered on the following wavelengths: 415, 753, 899, 952, and 1000 nm.

For the integrated analysis presented here, we focus on mineral and regolith parameters that directly influence broad spectral characteristics of the lunar regolith in the extended visible range accessible to CCD detectors. We specifically focus on the abundance and distribution of pyroxene as well as different compositions of pyroxene, the maturity of lunar soil (I_s/FeO), and the abundance of agglutinates. Our estimates of I_s/FeO will be shown to be highly correlated with the OMAT parameter of Lucey et al. (2000). Comparisons with the Lucey (2004) estimates of pyroxene distribution, however, can only be made in a qualitative manner since the Lucey approach is derived from radiative transfer modeled spectra that have been simplified for Clementine applications and that approach implicitly requires a large number of fundamentally different assumptions.

It is important to note that our analysis is strictly statistical by nature. The optical properties of exceptionally well characterized soils have been accurately measured, and we simply identify and use the most highly correlated relations. Such empirical studies are very popular and perhaps provide a sense of confidence, but they carry with them inherent dangers. They are naturally limited to the range of data included in the analysis and, more importantly, the uniqueness of their properties. It is now well known that the Apollo and Luna samples are not fully representative of the soil types across the Moon (e.g., Pieters, 1978; Lucey et al., 1998; Jolliff et al., 2000). In addition, several different compositional properties (e.g., strength of ferrous absorptions due to pyroxene, olivine, glass) are inherently linked and all contribute to the measured combined broad-band optical properties of low spectral resolution data. Furthermore, the statistics will be controlled by the most optically dominant material present in the collection of samples. And lastly, statistical correlations of optical properties with composition often exist not as a direct cause and effect, but due to a complex combination of poorly understood processes and products. A classic example is the correlation of blue-visible color with TiO_2 (Charette et al., 1974; Pieters, 1978; Johnson et al., 1991; Lucey et al., 1998; Gillis et al., 2003).

In the following sections, we present the results of our statistical analysis of the relation between lunar albedo and color as measured by Clementine with regolith parameters (I_s, I_s/FeO, and agglutinate abundance) and pyroxene composition for the most comprehensively studied suite of lunar soils (see Table 1). We provide detailed comparisons of prediction deviations and RMS errors. The RMS errors provide an estimate of the accuracy of the predictions, but only to the degree that this group of samples and these five Clementine spectral bands capture all the components of the diverse soil types found on the Moon. Because this is an optimized and self-contained statistical analysis, the RMS values are likely to be more optimistic than reality.

Therein lies the weakness of any statistical approach, of course. The physical basis for diagnostic mineral features is grounded in precise measurements with high spectral resolution data (e.g., Burns, 1993). High spectral resolution spectra are needed to both identify and characterize individual electronic transition mineral absorption bands. These highly diagnostic mineral absorption bands in the near-infrared are superimposed, especially near 1 μm for iron-bearing silicates, and require deconvolution approaches with high spectral resolution data for characterization (e.g., Sunshine et al., 1990; Sunshine and Pieters, 1993, 1998) or identification (e.g., Clark et al., 2003). When the number of spectral bands are limited, there is insufficient spectral resolution for a unique solution. Thus, an analysis using multispectral images can only provide first-order assessment, regardless of the approach. The comparative analysis of optimized formulations using Clementine multispectral images presented here illustrates how a statistically accurate structure produces spatially coherent results that are tempting to interpret in detail (but sometimes contradictory). The combined results provide useful insights about the first-order distribution of materials across the Moon, but the user should always be cautious about applying simple algorithms that depend on short cuts using multispectral data results should not be over-interpreted.

3. Approach: Compositional predictions from ground truth

Our overall approach is shown as a flow chart in Fig. 2 and is discussed below. The goal of our approach is to establish the best mathematical link between independent data. We search for the closest correlation between a specific compositional pa-

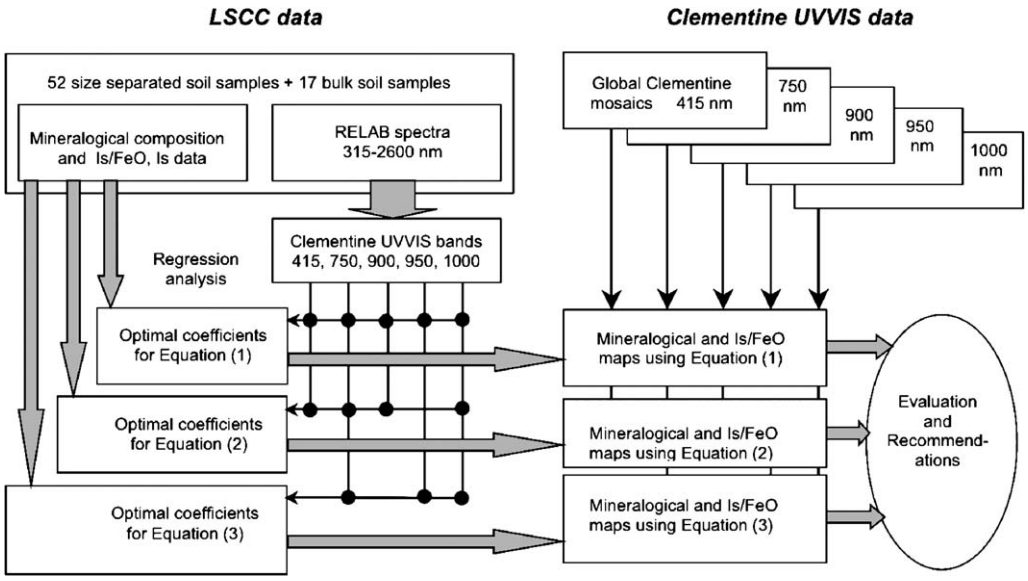


Fig. 2. Flow chart illustrating the approach used. See text for discussion.

Table 2
Coefficients of Eq. (1) found using data for samples from Table 1, RMS deviation (σ), and correlation coefficient (k)

	a_1	a_2	a_3	a_4	a_5	a_6	a_7	σ (%)	k
Total Px	−0.021	−0.033	8.401	−24.521	12.981	−3.102	7.311	4.43	0.88
Pigeonite	−0.021	−1.182	11.911	−27.716	13.205	−2.375	6.260	2.04	0.84
Augite	−0.056	−1.211	57.985	−30.517	−9.890	12.611	−28.296	3.59	0.82
OPx	−0.003	0.425	−26.464	−22.127	32.838	−11.906	26.985	1.44	0.80
Fe-Px	−0.032	5.550	−8.677	−62.386	51.859	−19.963	35.191	1.82	0.45
I_s/FeO	−0.014	−1.892	−20.811	16.971	−1.679	−5.124	13.932	23.41	0.88
Agg.	−0.004	−1.020	2.713	−0.404	−0.847	1.006	−0.103	7.45	0.82
I_s	−0.003	−1.813	−1.772	2.067	−0.761	−2.085	6.940	196.91	0.92

rameter, P, and an empirical combination of spectral albedo, A, of the lunar samples in the 5 Clementine bands. This involves optimizing a set of coefficients to minimize the RMS error between the observed and the predicted values of P. Once the best set of coefficients is derived from the combined laboratory data, we apply the formula to map the parameter using Clementine data and then evaluate the results.

This approach depends in part on the choice of empirical combinations of spectral bands, and there is no unique formulation of optical parameters. We have explored several options, most of which are derived from an understanding of the causes of spectral variations (e.g., ferrous absorption near 1 μm, continuum and albedo change with exposure, etc.). No single option is perfect, but we provide several examples of the statistically best. General requirements sought are that the produced parameter maps: (1) are spatially coherent (with small local deviations); (2) do not contain major latitude trends (likely related to residual photometric errors of the Clementine mosaics); (3)do not contain spatially coherent negative values of the predicted parameters; and (4) provide the highest correlation coefficients between predicted and measured parameters. Although we tested multiple combinations, we use and compare the applicability of three formulations here that turned out to provide the best results. In all cases we used $\log P$ in the formulation in order to avoid negative predicted values.

The first equation is similar to the one used in our previous investigation (Shkuratov et al., 2003a). It includes one albedo term and several spectral ratios:

$$\log P = a_1 A_R + a_2 C_{\mathrm{BR}} + a_3 C_{\mathrm{IR1}} + a_4 C_{\mathrm{IR2}} + a_5 C_{\mathrm{IR3}} + a_6 D + a_7, \quad (1)$$

where P is the studied parameter, the coefficients a_i ($i = 1, \ldots, 7$) are chosen to minimize the RMS deviations of the calculated values of P from the measured values. The optical terms are: albedo $A_R = A(750\ \mathrm{nm})$ [%], color-indexes: $C_{\mathrm{BR}} = A(415\ \mathrm{nm})/A(750\ \mathrm{nm})$, $C_{\mathrm{IR1}} = A(900\ \mathrm{nm})/A(750\ \mathrm{nm})$, $C_{\mathrm{IR2}} = A(950\ \mathrm{nm})/A(750\ \mathrm{nm})$, $C_{\mathrm{IR3}} = A(1000\ \mathrm{nm})/A(750\ \mathrm{nm})$, and bend $D = A(750\ \mathrm{nm})A(1000\ \mathrm{nm})/[A(900\ \mathrm{nm})]^2$. It was expected that the three IRn color indexes and the bend ratio would capture the principal variations of the ferrous absorption near 1 μm. Albedo coupled with any of these color ratios should capture effects of space weathering (e.g., Lucey et al., 2000; Staid and Pieters, 2000).

The second equation used is the simplest linear combination of albedo in the principal spectral bands (A in %).

$$\log P = b_1 A_{415} + b_2 A_{750} + b_3 A_{900} + b_4 A_{1000} + b_5. \quad (2)$$

Table 3
Coefficients of Eq. (2) found using data for samples from Table 1, RMS deviation (σ), and correlation coefficient (k)

	b_1	b_2	b_3	b_4	b_5	σ (%)	k
Total Px	−0.067	0.221	−0.027	−0.163	1.332	4.70	0.85
Pigeonite	−0.142	0.309	0.051	−0.276	0.856	2.23	0.84
Augite	−0.191	0.211	0.616	−0.727	1.206	1.60	0.93
OPx	−0.039	0.320	−0.784	0.469	0.271	7.32	0.51
Fe-Px	0.238	0.127	−0.554	0.226	0.573	3.93	0.25
I_s/FeO	−0.0012	−0.219	0.097	0.099	1.945	17.42	0.92
Agg.	−0.058	0.013	0.058	−0.037	1.753	6.51	0.85
I_s	−0.032	−0.163	0.204	−0.055	3.240	212.28	0.92

Table 4
Coefficients of Eq. (3) found using data for samples from Table 1, RMS deviation (σ), and correlation coefficient (k)

	c_1	c_2	c_3	c_4	σ (%)	k
Total Px	0.123	−0.259	0.111	1.365	5.07	0.80
Pigeonite	0.128	−0.348	0.187	0.955	2.47	0.72
Augite	0.069	0.145	−0.272	1.401	1.71	0.85
OPx	0.115	−1.020	0.877	0.227	6.51	0.52
Fe-Px	0.347	−0.306	−0.059	0.352	5.30	0.28
I_s/FeO	−0.200	0.161	0.018	1.960	17.50	0.92
Agg.	−0.057	−0.064	0.108	1.803	7.37	0.81
I_s	−0.168	0.161	−0.033	3.280	208.20	0.92

This equation captures variations in the visible as well as those near the 1 μm ferrous band. Preliminary results have been presented in Shkuratov et al. (2003b) and Omelchenko et al. (2003).

The third equation is an even more simplified form, and preliminary results have been briefly discussed (Shkuratov et al., 2003a; Kaydash et al., 2004),

$$\log P = c_1 A_{750} + c_2 A_{950} + c_3 A_{1000} + c_4, \quad (3)$$

where A is given in %. In this formulation we exclude the shortest wavelength band and substitute the 950 nm albedo for the 900 nm albedo. This formulation should thus be less sensitive to overall continuum variations.

In Tables 2–4 we provide the derived coefficients for Eqs. (1)–(3) for each of eight lunar soil characteristics (parameters, P) that were accurately measured by the LSCC: the abundance of total pyroxene, the abundances of four different compositions of pyroxenes (orthopyroxene, pigeonite, augite, and Fe-rich pyroxene), agglutinate abundance, and parameters associated with the degree of maturity (ferromagnetic resonance response, I_s, and I_s/FeO; see Morris, 1976, 1978, for a discussion of these parameters). In addition, the tables include the correlation coefficients k and RMS deviation σ that characterize correlation between the measured (by LSCC) and predicted values (through the relations (1)–(3)) of each parameters. The statistics for each compositional and maturity parameter is analyzed separately. Thus, although agglutinates tend to increase as lithic fragments are consumed, the % of either parameter in the soil (agglutinate glass, pyroxene, etc.) is treated independently. We note once again that the listed correlation coefficients, k, are found after minimizing the RMS deviation of predicted values from the measured values.

4. Results

No single formulation is statistically best for all parameters. It should be emphasized that the correlation coefficient between predicted and measured values characterizes not only the "quality" of correlation, but mathematically it is also a function of the number of linear combination coefficients used in equations such as Eqs. (1)–(3). For example, if the number of the linear combination coefficients equals the number of samples, the RMS minimization approach will cause the correlation coefficients to approach 1.0 for the samples. Thus, if Eqs. (1) and (3) provide equal correlation coefficients, one should generally prefer Eq. (3) since it has a smaller number of terms.

Comparison of the tables shows that Eq. (1) or Eq. (2) provide, as expected, generally higher correlation coefficients than Eq. (3). However, for the agglutinate content, I_s, and I_s/FeO, the simpler Eq. (3) gives high correlation coefficients that sometimes are even larger than for Eq. (1) or Eq. (2). For the mineral parameters (various types of pyroxene), however, Eq. (1) or Eq. (2) are always better than Eq. (3). For orthopyroxene, only one formulation, Eq. (1), provides a statistically attractive correlation. Values for Eq. (1) or Eq. (2) are similar for total pyroxene and pigeonite, but Eq. (2) is distinctly better for augite and Fe-pyroxenes. Since Eq. (2) has 5 terms, whereas Eq. (1) deals with 7 terms, we tend to prefer Eq. (2) for all mafic minerals except orthopyroxene. Clearly, increasing the number of samples used in the analysis provides greater confidence in capturing the diversity inherent in the samples, but does not directly affect the accuracy with which properties can be captured by the number of terms.

Comparisons between the measured values (by LSCC) and the predicted values of the parameters (derived from the three different formulations discussed above) are shown in Fig. 3. For each compositional and maturity parameter there are three

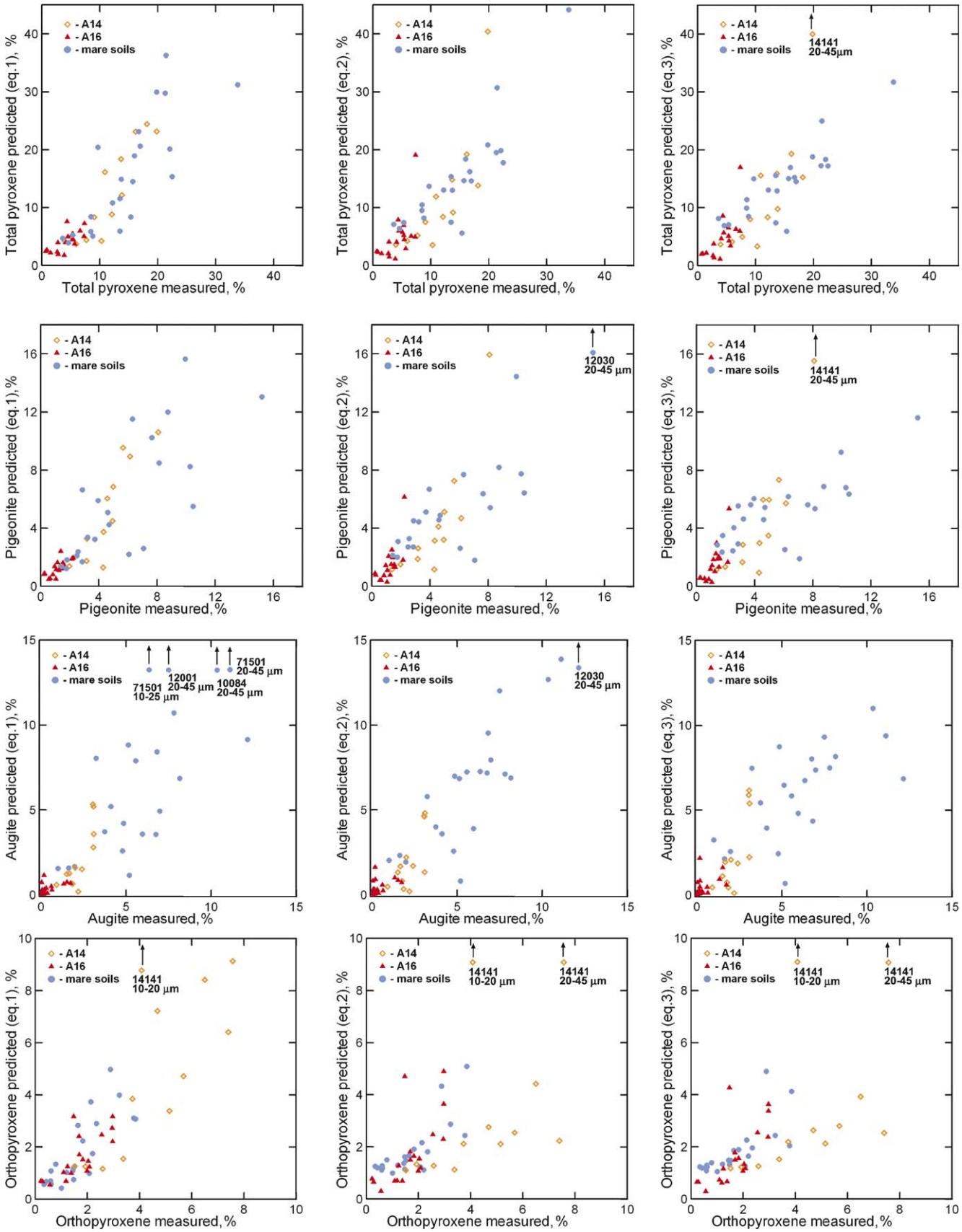


Fig. 3. Scatterplots of measured and best fit predicted values for compositional parameters using data from Table 1. Each row is a different parameter. Values are shown using Eq. (1) [left], Eq. (2) [middle], and Eq. (3) [right]. Three particle sizes of soil separates are used for derivation of coefficients. The figures for I_s/FeO and I_s, contain additional data for bulk samples.

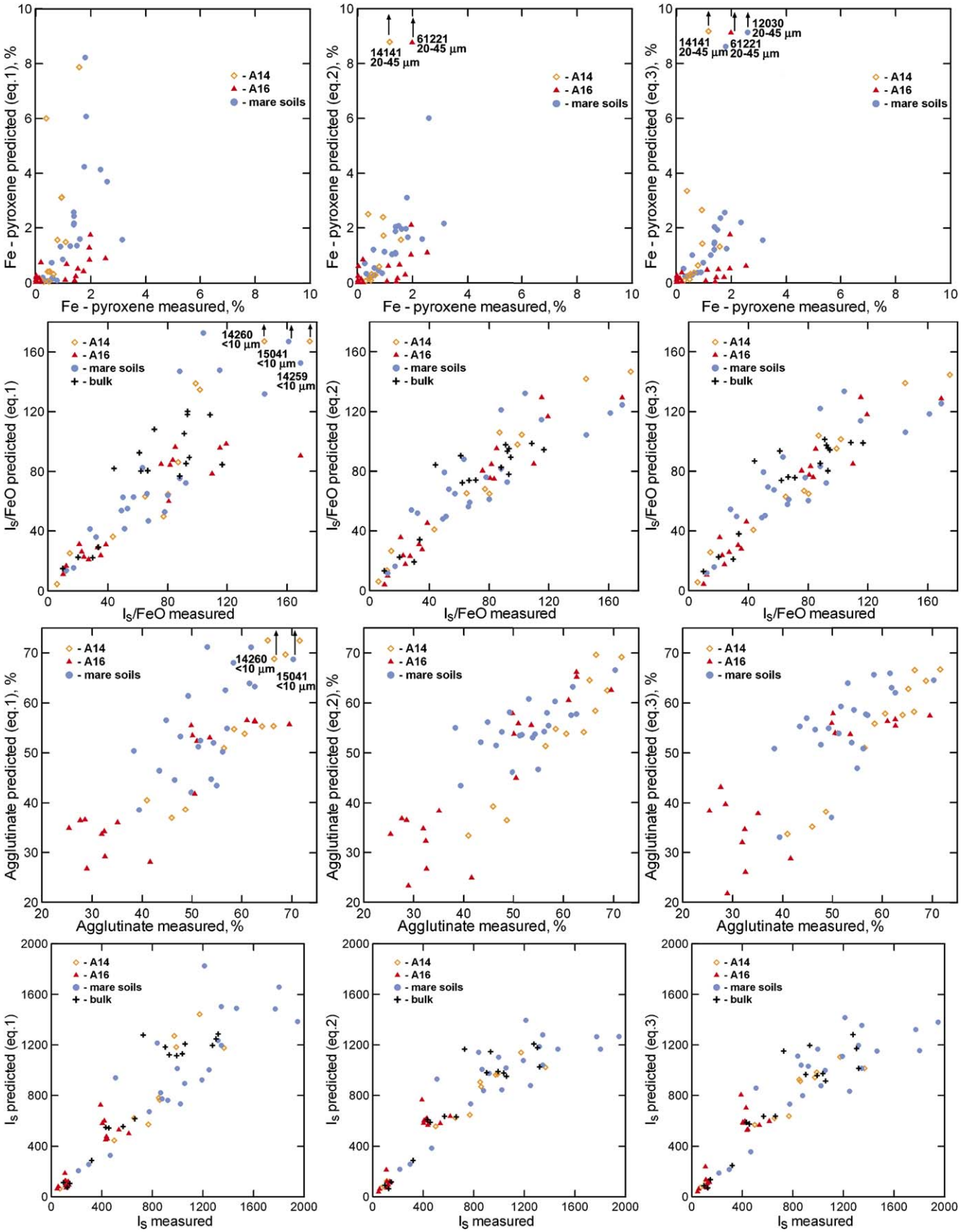


Fig. 3. (*continued*)

plots, one for each of the three equations used. Predicted values that are beyond the range of the plot are indicated at the top of each figure. For mineral parameters, compositional data is available only for the three soil size separates. On the other hand, major element chemistry and I_s values are available for both the size separates as well as for the bulk sample (e.g., see Taylor et al., 2001, and LSCC website). For I_s and I_s/FeO, the coefficients shown in Tables 2–4 were derived using only the LSCC values for the three particle-size separates of each soil. We then use the independently measured chemistry and I_s values for the bulk <45 μm sample for each soil in order to test the quality of our prediction for these two parameters. The predicted values for these bulk samples are nevertheless included on the plot even though they were not involved in the RMS calculations. Data for the independent bulk samples in Fig. 3 fall in the middle of the data for the size separates used to derive the coefficients for Tables 2–4, confirming statistical trends of the data.

5. Assessment of precision and quality

How well can the equations predict compositional parameters across the lunar surface and which formulation is best? This is difficult to evaluate since we have used the "ground truth" data available for lunar soils to derive the formulations.

The first step is to assess the correlation coefficients of the predicted and measured data, k, and the standard deviation, σ, of these values from a perfect 1:1 relation as summarized in Tables 2–4. Note that k is calculated through a logarithmic relation and σ is derived from a linear relation. We set a correlation coefficient of 0.50 as a lower threshold for a meaningful relationship between properties. The following observations can be made:

1. None of the formulations are adequate to accurately predict the amount of iron-rich pyroxenes [Fe-Px] in lunar soil. This is not surprising since the actual measured abundances of pyroxene with such a composition are very low and the relative effect of iron-rich pyroxene on the optical properties is small.
2. For orthopyroxene [OPx], Eq. (1) is substantially better than the other two as measured by both k and σ.
3. For augites, Eq. (2) is substantially better than the other two as measured by both k and σ.
4. For pigeonite [Pigeo], Eqs. (1) and (2) are both notably better than Eq. (3) as measured by both k and σ. There is no clear preference for one or the other, however.
5. For total pyroxene [Total Px], there is a general improvement in k and σ as the number of terms in the equation increases, but as discussed above, this is to be expected. Since all values of k and σ are reasonably good, there is thus no clear preference between the equations.
6. For I_s there is a high k for all formulations with little variation in σ. A preference is thus for Eq. (3) since it has the fewest terms.
7. I_s/FeO has excellent values of k and σ for both Eqs. (2) and (3). Since I_s appears to have a preference for Eq. (3), it is reasonable to expect the same formulation for the related I_s/FeO.
8. Agglutinates [Agg] appear to be best estimated with Eq. (2), but it is not clear whether Eq. (3) is equally appropriate since its high correlation coefficient is derived from the fewest terms.

Table 5
Recommended preferences in the choice of equations for each parameter

	Eq. (1)	Eq. (2)	Eq. (3)
Total Px	?	√	?
Pigeonite	?	?	
Augite		√	
OPx	√	?	
Fe-Px			
I_s/FeO			√
Agg.		?	√
I_s			√

√ is moderately preferred; ? is ambiguous.

Inspection of these results allow a few ambiguities to be removed. For the pyroxenes, none of the individual pyroxene compositions favor the simple Eq. (3); all are best described by Eq. (1) or Eq. (2). Thus, Eq. (3) should also be eliminated as a good option for total pyroxene. Also, the relations seen in Fig. 3 for total pyroxene indicate that Eq. (1) produces somewhat non-linear results for values of total pyroxene abundance. These observations suggest that Eq. (2) should be preferred for total pyroxene estimation. These observations about preferred formulation for predicting values of different compositional and maturity parameters are summarized in Table 5.

6. Estimates using Clementine data

In order to use these results with Clementine data, it is necessary to evaluate the spatial extent of materials predicted and to look for spatial coherency or artifacts that may jeopardize the validity of any predictions. We initially use the 1-km Clementine UV–VIS mosaics re-sampled to a resolution of 15 km for global display purposes. A Clementine 750 nm albedo image is shown in Fig. 4 as an overview. Figs. 5–9 are derived maps that predict the distribution of different types of pyroxene; the upper, middle, and lower panels of each figures presents results obtained with Eqs. (1)–(3), respectively. Figs. 10–12 show similar maps of the maturity-related parameters. For ease of comparison, the same scale is used for each of the results from the three different formulations.

Even though the three equations are quite different, all of the derived maps in Figs. 5–12 provide spatially coherent values. In general, spatially coherent maps would be produced by almost any formulation of spectral parameters that are coupled to spectral variance, provided that the inherent signal-to-noise ratio of the input data is not exceeded. For each parameter the three maps are similar, indicating that the approach is relatively stable. There are several observations that can be made about both the content and diversity seen in these compositional and maturity maps:

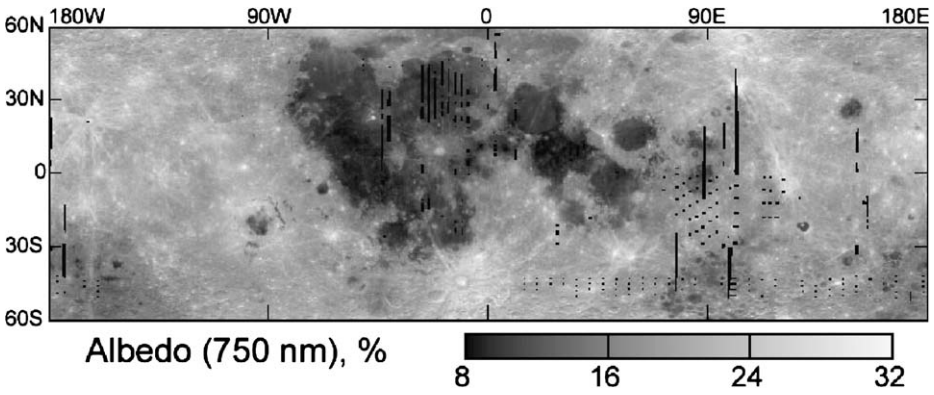


Fig. 4. Clementine 750 nm albedo mosaic.

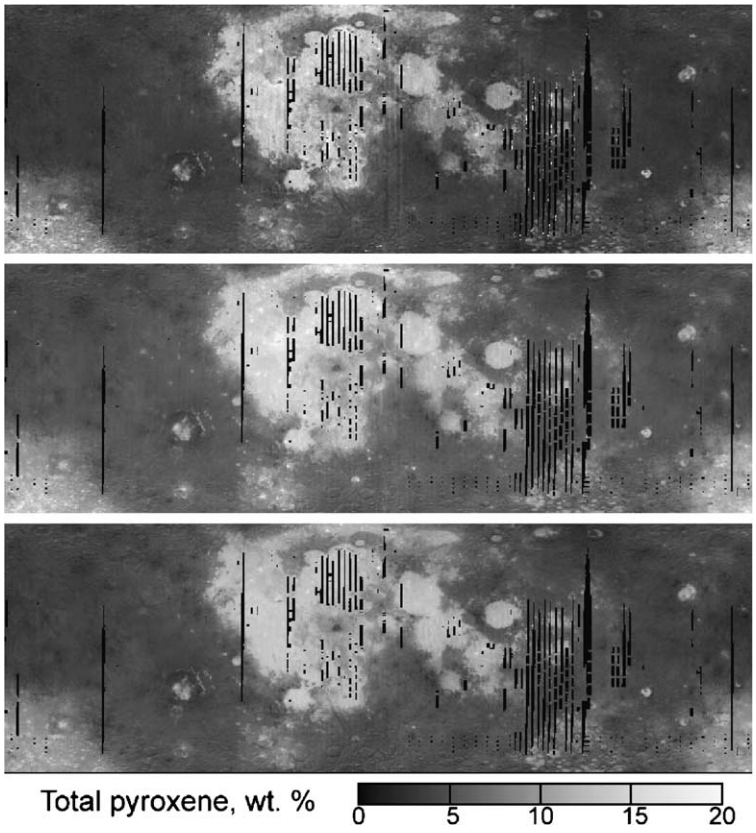


Fig. 5. Estimates of total pyroxene abundance with 15 km resolution. The upper, middle, and lower panels correspond to Eqs. (1)–(3), respectively.

a) As expected, the lunar maria are predicted to have a high abundance of pyroxene and the highlands to have a low abundance.
b) Maps of the two maturity parameters I_s/FeO and agglutinates are very similar since these components are both the result of exposure to the space environment. Both parameters readily map material freshly exposed by a recent impact crater, including extensive ray systems. Because almost no mare/highland differences are observed in these I_s/FeO and agglutinate distribution maps, the parameters appear relatively insensible to compositional variations.
c) The distribution maps of I_s reflect the abundance of nanophase metallic iron (npFe^0) produced during the soil formation process (Pieters et al., 2000; Hapke, 2001; Noble et al., 2004, 2001). This parameter is dependent on both the amount of Fe available in the host material and the length of time of exposure. The patterns of fresh craters (low) and Fe-rich mare (high) are expected.
d) Fresh craters in the maria have the highest abundance of pyroxene, particularly augite and pigeonite. Again, this is expected due to the proportional decrease in lithic components and increase in agglutinitic glass as soils mature (e.g., Taylor et al., 2001). (See also the discussion of higher resolution data below.)
e) Some, but not all, fresh craters in the highlands exhibit an enhanced pyroxene concentration at the crater itself (but typically not the rays). Other highland craters and several massifs of basins (Humorum, Orientale, Nectaris) are no-

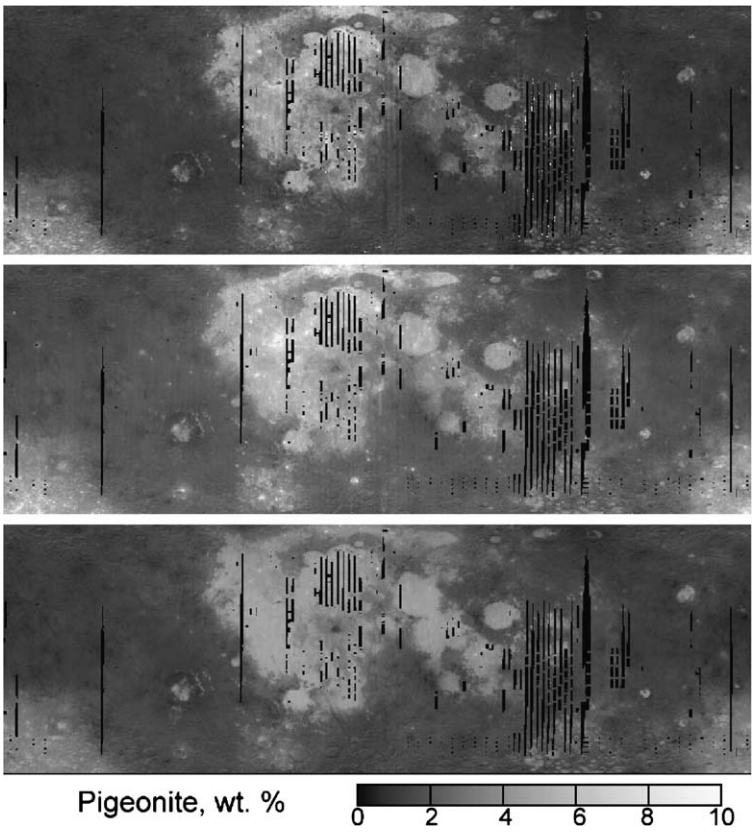


Fig. 6. Estimates of pigeonite abundance with 15 km resolution. The upper, middle, and lower panels correspond to Eqs. (1)–(3), respectively.

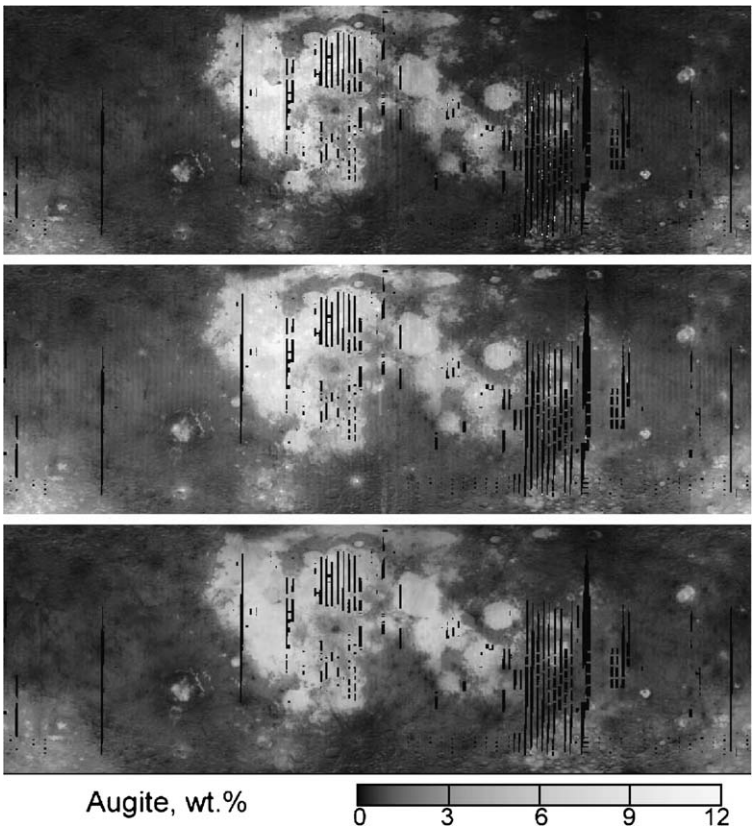


Fig. 7. Estimates of augite pyroxene abundance with 15 km resolution. The upper, middle, and lower panels correspond to Eqs. (1)–(3), respectively.

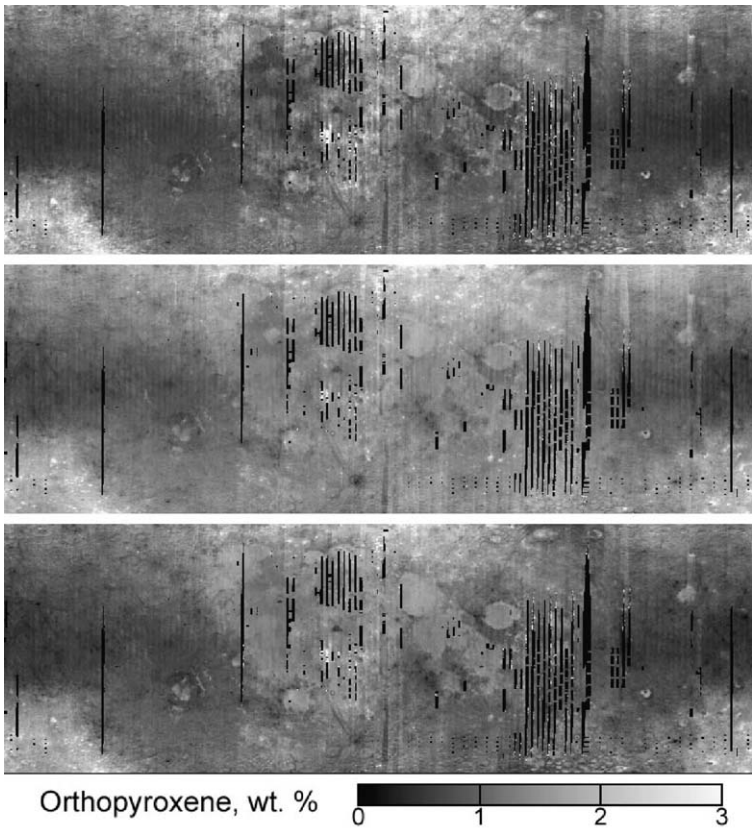


Fig. 8. Estimates of orthopyroxene with 15 km resolution: the upper, middle, and lower panels correspond to Eqs. (1)–(3), respectively.

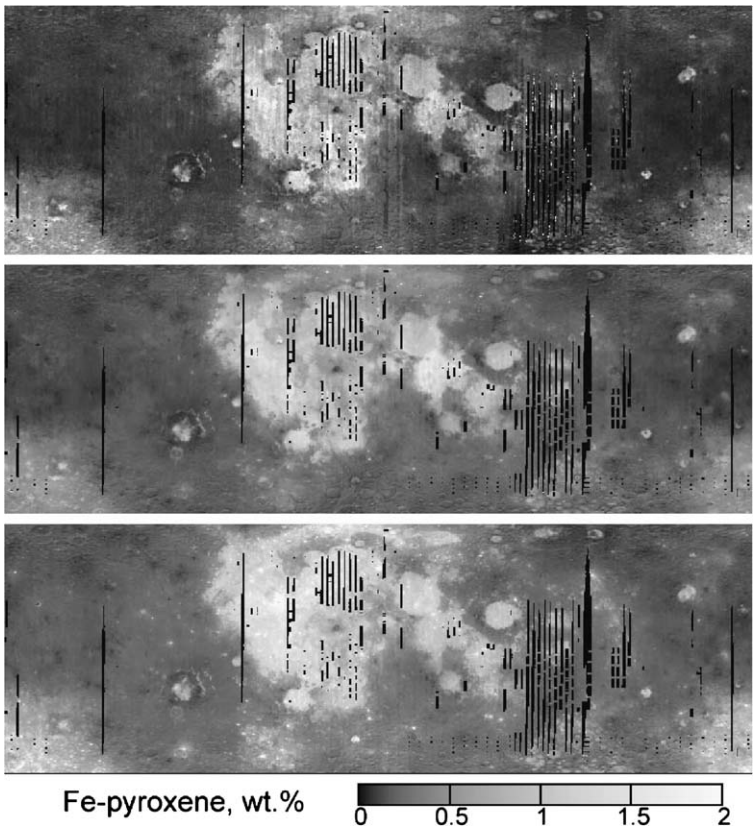


Fig. 9. Estimates of Fe-rich pyroxene abundance with 15 km resolution. The upper, middle, and lower panels correspond to Eqs. (1)–(3), respectively.

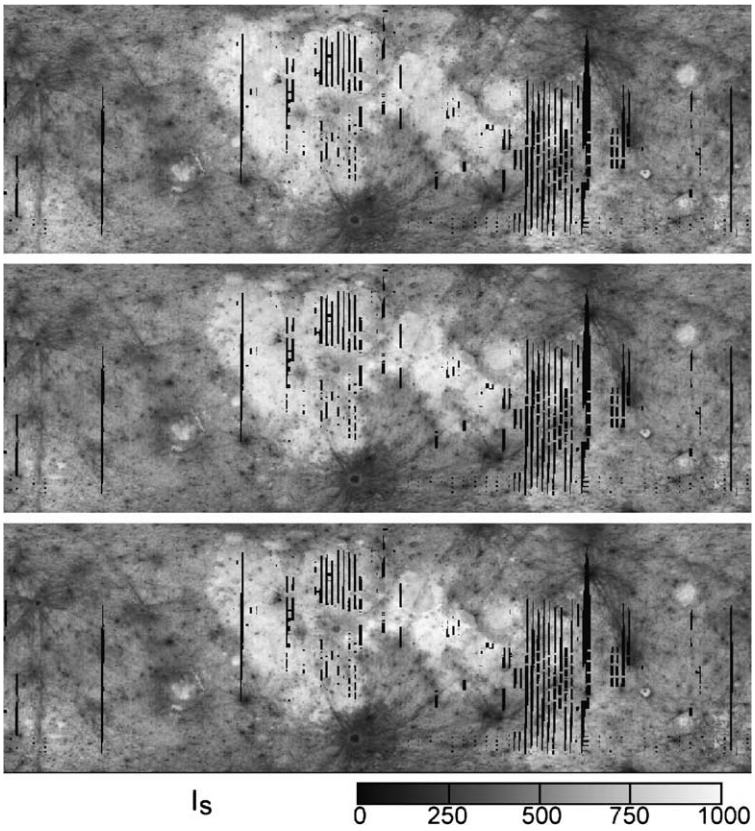


Fig. 10. Estimates of I_s parameter with 15 km resolution. The upper, middle, and lower panels correspond to Eqs. (1)–(3), respectively.

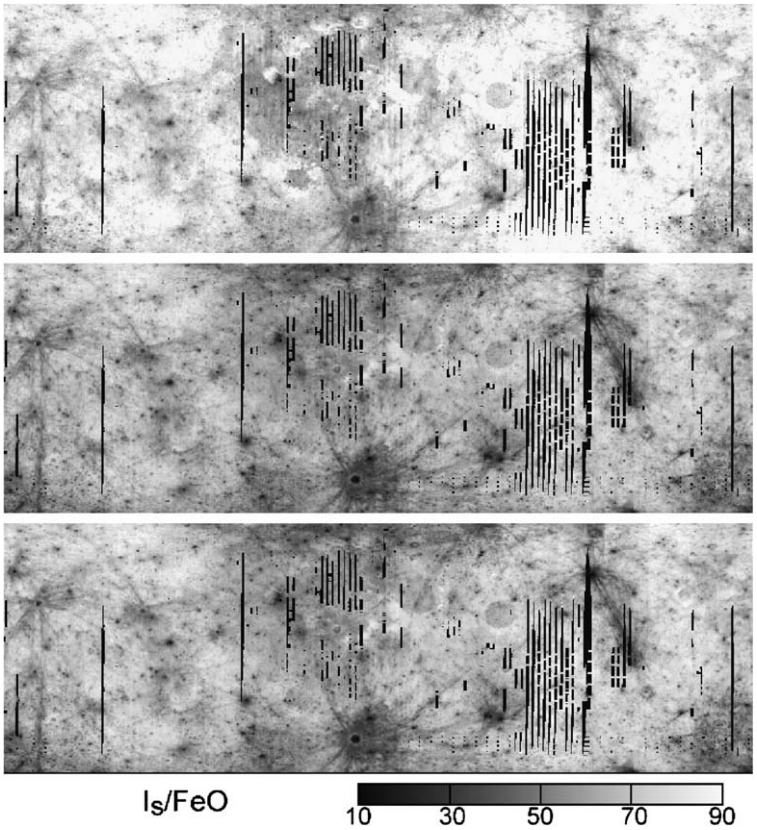


Fig. 11. Estimates of the maturity degree I_s/FeO with 15 km resolution. The upper, middle, and lower panels correspond to Eqs. (1)–(3), respectively.

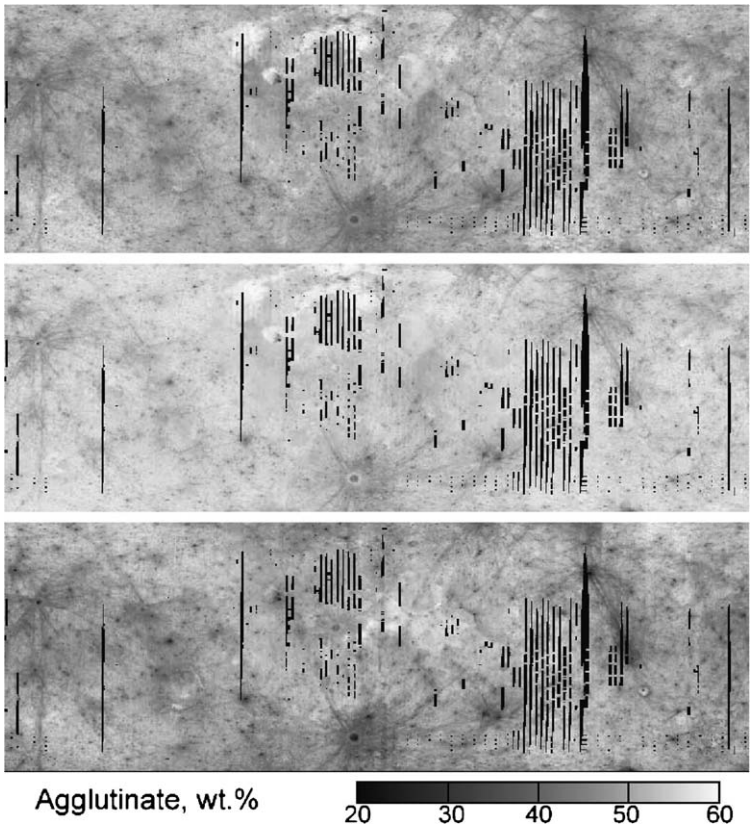


Fig. 12. Estimates of agglutinates abundance with 15 km resolution. The upper, middle, and lower panels correspond to Eqs. (1)–(3), respectively.

tably devoid of pyroxene, consistent with an interpretation of exposed anorthosite (e.g., Hawke et al., 2003).

f) All three maps for orthopyroxene exhibit a distribution that contains an apparently latitude dependent component, most prominently seen on the farside. Although the farside equatorial highlands could conceivably be unusually devoid of orthopyroxene, a possible explanation is that this parameter is particularly sensitive to residual wavelength dependent photometric calibration shortcomings of the Clementine data.

g) In spite of the potential residual calibration issues, an unusual concentration of orthopyroxene is seen across South Pole–Aitken basin as well as the highlands north of Imbrium (Pieters et al., 2001; Pieters, 2002).

h) Relative values of agglutinate content between Mare Serenitatis and Mare Tranquillitatis, however, are observed to differ depending on whether we used Eq. (2) or Eq. (3). The former suggests that the Mare Tranquillitatis regolith contains less agglutinates than that of Mare Serenitatis. The opposite is observed for the latter, Eq. (3). Differences in well developed regolith might be linked to age or composition effects. In this example, it is difficult to discern which prediction is more likely to represent reality, and interpretations of such broad differences remain ambiguous.

In summary, the formulation for a suite of mineral and maturity parameters appear to produce global results that are consistent with known relations between different lunar materials. It should be noted, however, that these are general trends. The specific details and actual predicted values depend on the formulation used. Since the global data available consist of only 5 bands between 400 and 1000 nm, the predictions for compositional and maturity values are only general estimations. Recall that the error associated with k and σ are *best case* statistical fits for the specific group of samples analyzed by LSCC. There is no good way to evaluate the validity with high precision for unsampled areas with this limited data.

The discussion above is related to global Clementine maps. Evaluation of the parameters at higher resolution, e.g., 1 km per pixel, is more appropriate for analyses of regional geology. A comparison between pyroxene estimation maps derived using Eqs. (1) and (2) (see Table 5) is shown in Figs. 13 and 14 for a small region of 1-km data centered on the Crisium basin. The broad similarity observed globally for pyroxene maps derived with different equations begins to deteriorate in higher spatial resolution data. In particular, all estimates derived using Eq. (1) result in maps that exhibit significantly lower effective spatial resolution than those derived using Eq. (2). This magnification of cumulative noise that appears in Eq. (1) results is a common property of equations with a high number of terms. Thus, for practical purposes we generally recommend use of Eq. (2) even if its statistical parameters, k and σ, are worse than those of Eq. (1).

Several interesting features can be seen in the higher resolution maps of Fig. 14 derived from Eq. (2). For example, the crater Picard in SE Crisium appears to exhibit faint “rays”

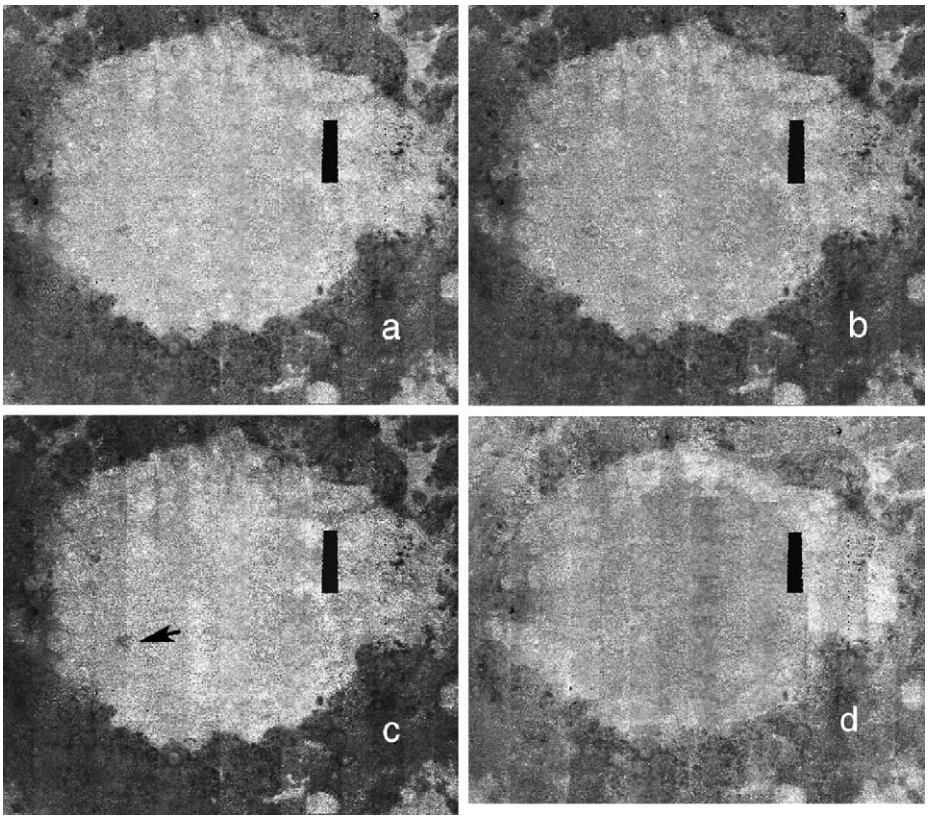


Fig. 13. Estimates of pyroxene abundance for the Crisium basin using 1-km resolution images derived using Eq. (1). Panels (a)–(d) correspond to total pyroxene, pigeonite, augite, and orthopyroxene, respectively. The crater Picard is shown with an arrow.

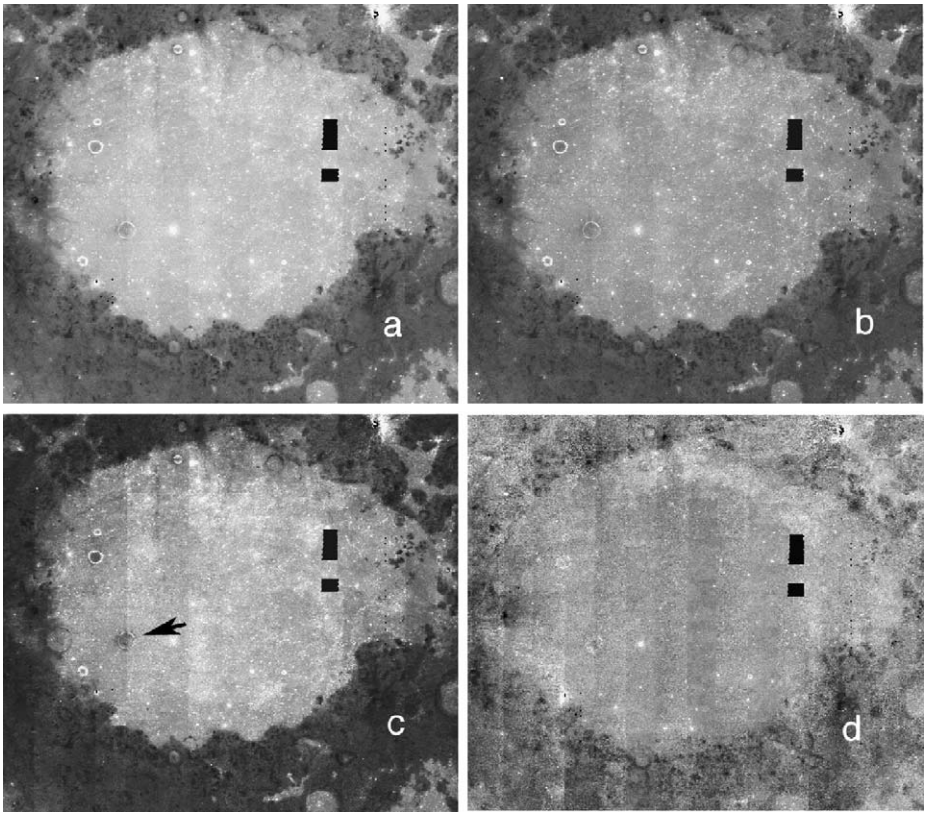


Fig. 14. Estimates of pyroxene abundance for the Crisium basin using 1-km resolution images derived using Eq. (2). Panels (a)–(d) correspond to total pyroxene, pigeonite, augite, and orthopyroxene, respectively. The crater Picard is shown with an arrow.

of low pyroxene abundance emanating up to a crater radius from the rim. Could these be glass-rich areas or do the unusual rays suggest a small amount of highland material has been tapped by the crater? Previous analyses of this crater have noted that it appears to have excavated basaltic material of a different composition from the surroundings (Andre et al., 1978; Head et al., 1978). Perhaps one of the basalts has a distinctly lower pyroxene abundance. There is also a notable suggestion of enhanced orthopyroxene distribution forming a ring near the periphery of Mare Crisium. Although neither of these observations can be validated or clarified with the data in hand, they highlight interesting areas to be targeted with the next generation of instruments to be flown to the Moon that are more sensitive to lunar mineralogy (higher spectral resolution and a broader spectral range into the near-infrared).

7. Data comparisons

It is also important to test our parameter predictions and results for internal consistency and to compare results with similar parameters described elsewhere. Shown in Fig. 15 are trends between several compositional and maturity parameters. The LSCC measured values are superimposed on the global Clementine predicted values. We also include examples to illustrate relations between our data and the Lucey et al. (2000) OMAT maturity parameter.

The estimated parameter values from Clementine data are generally well bounded by the actual LSCC measured values in Fig. 15. For augite pyroxene there is a distinct inflection in the Clementine data for higher pyroxene values (see Fig. 15A). This appears to be tied to the higher abundance of this high-Ca pyroxene for mare basaltic regions. Similarly, the LSCC data clearly illustrate the preponderance of orthopyroxene in Apollo 14 and 16 soils with a presence, but lower abundance, in mare soils (see Fig. 15B). The distribution of estimated values from Clementine data appears to mirror this affinity for orthopyroxene in the more ancient feldspathic highlands.

On the other hand, the relation between I_s/FeO and agglutinate content appears to be bi-modal (Fig. 15C). The principal relation between I_s/FeO and agglutinate distributions reflects the process of increasing the amount of nanophase iron relative to total iron in the volume of regolith particles during agglutination, a direct function of soil maturity. The two main clusters of Fig. 15C correspond to the maria and highlands. This structure suggests there might be some process linked to the inherent difference in composition in mare and highlands that distinguishes these two soil maturity parameters (perhaps the proportion of meteoritic I_s contribution would be greater in the low-Fe highlands, skewing the I_s/FeO). Craters and their ray systems form the tails of the clusters.

In order to compare our parameters that estimate maturity of the lunar surface to the OMAT parameter described by Lucey et al. (2000), we need to examine the LSCC data in the same framework that defines the OMAT parameter. Shown in Fig. 15D are values for the LSCC soils comparing 750 nm albedo with a color ratio that is sensitive to the strength of ferrous absorptions in minerals. The Lucey et al. (2000) OMAT parameter is essentially the distance of any given point to their empirically defined origin (marked with an "x" in Fig. 15D), with greater distance being more mature. The equation for determination of OMAT is:

$$\text{OMAT} = \sqrt{\big(A(750) - x\big)^2 + \big[\big(A(950)/A(750)\big) - y\big]^2}, \qquad (4)$$

where $x = 0.08$, $y = 1.19$ (Lucey et al., 2000). As with the compositional parameters discussed above, the LSCC data tend to bound the values for Clementine data for the two optical parameters shown in Fig. 15D. One note of concern, however, is that the particle-size trends of LSCC data (which are known to contain increasing amount of agglutinates and I_s/FeO with decreasing particle size) do not follow a consistent relation in this type of representation. In particular, the smallest size fraction for some Apollo 16 soils are further from the Lucey et al. (2000) "origin" than the larger size fraction, whereas the opposite is true for most mare soils.

Shown in Figs. 15E and 15F are comparisons of agglutinate estimates for Clementine 1-km data derived from Eqs. (2) and (3) with values for the OMAT maturity parameter of Lucey et al. (2000). As seen in Fig. 15E there is a correlation between the two for agglutinate estimates using Eq. (3). This is, of course, to be expected since both formulations rely on a similar and very limited number of Clementine bands. On the other hand, agglutinate abundance estimated for Clementine data using Eq. (2), which also includes visible wavelengths and has a higher correlation coefficient, is equally well bound by the LSCC data (Fig. 15F). In this case, the more extended pattern between the two independently derived optical parameters underlines the fact that they are not identical. Although the strong correlation seen in Fig. 15E is attractive, to be cautious, we still have no independent way of knowing whether the formulations for OMAT and this formulation for agglutinate abundance are both relatively valid or both equally inaccurate. They are correlated because they were designed that way; both use the same range of spectral information.

8. Conclusions

- We have provided a statistical evaluation of relations between mineralogy and spectral parameters for lunar soils extracted from 5-band multispectral image data from Clementine.
- Pyroxene mineralogy and maturity trends can be identified and mapped using Clementine data that are consistent both with the measured properties of lunar soils as well as expectations for global properties. These provide a valuable first-order assessment of bulk mineralogy.
- However, deceivingly coherent maps can also be produced even when the predicted and measured data in fact show no inherent relation (e.g., Fe-pyroxene estimates). Producing a spatially coherent map from any simple mathematical parameter that captures spectral variance only means the input data have low noise. We caution against the impulse for over-interpretation.
- A constructive use of these results is to identify unusual areas that merit further investigation with more advanced

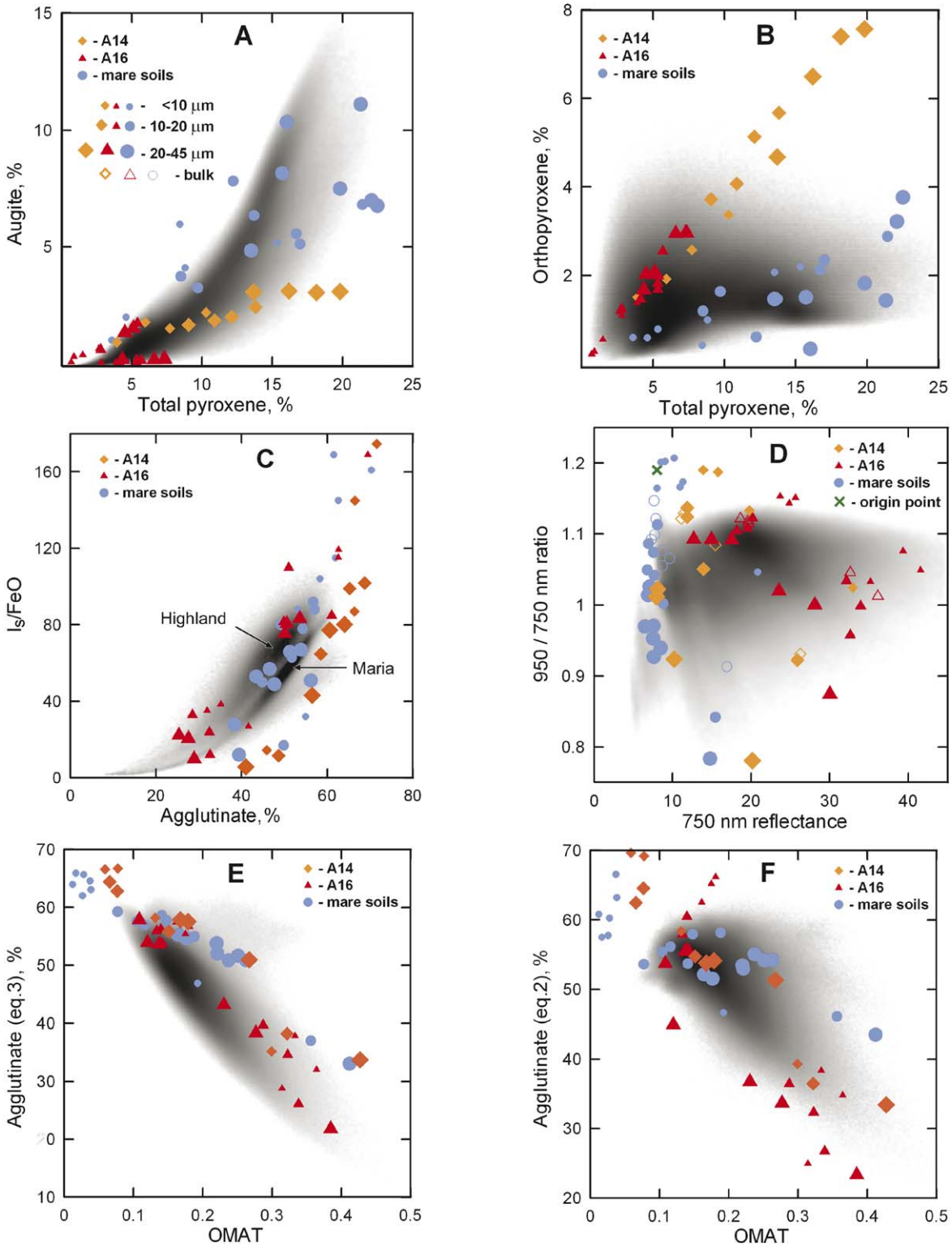


Fig. 15. Relation between values measured in the laboratory for LSCC soils (symbols) to values derived from Clementine 1-km data. The equations used for derivation of the compositional parameters shown are: (A) Eq. (2); (B) Eq. (2); (C) Eq. (3); (E) Eq. (3); (F) Eq. (2). OMAT is derived according to Lucey et al. (2000). The values of optical parameters used by Lucey et al. (2000) as an origin for derivation of OMAT is indicated with an X in (D).

instruments designed to characterize mineralogy with data of high spectral resolution and broad spectral coverage. Near-infrared spectrometers that are flown on the current group of lunar orbital missions (Chandrayaan-1, SELENE) will provide the first detailed assessment of lunar mineralogy and related resources on a global, regional, and local scale for the next generation of geologic analyses of the Moon (e.g., Pieters et al., 2006).

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